THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 08

Full Syllabus (Chapters 1–13) | Class 9 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** ask what mechanism and measurable change could make it harmful	test claims with questions and evidence
2	**(b)** it is far too little for a whole month	sense check
3	**(b)** oil and water	form separate layers
4	**(d)** 9 %	$\frac{18}{200} \times 100 = 9$
5	**(b)** MgO	Mg^{2+} and O^{2-}
6	**(a)** sulfur hexafluoride	prefixes show the number of atoms; second element ends in -ide
7	**(a)** most of the atom is empty space	Rutherford's inference
8	**(d)** 25 m	$v^2 = u^2 + 2as$
9	**(b)** 30 m s^{-1}	108 × 5/18
10	**(a)** 20 kg	m = F/a
11	**(b)** Newton's third law	equal and opposite force
12	**(b)** 250 W	P = W/t
13	**(c)** 0.1 s	persistence of hearing
14	**(b)** genes	units of heredity
15	**(b)** increase in the number of cells	cell division
16	**(c)** apical meristem	tips
17	**(b)** air quality	colour changes with pollutants
18	**(c)** yeast and hydra	outgrowth
19	**(d)** placenta	umbilical cord
20	**(d)** troposphere	lowest layer

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	Physics: pressure–temperature relation; chemistry: cooking changes food chemically (or biology: nutrients) (1+1).	1+1
22	CO_2: $12 + 32 = 44$ u; NH_3: $14 + 3 = 17$ u (1+1).	1+1
23	Displacement (1); rectangle: velocity × time (1).	1+1
24	It increases the time to stop the ball, so the acceleration and force are smaller (2).	2
25	Cellulose (1); roughage that aids digestion (1).	1+1
26	Parenchyma: loosely packed living cells that store food; collenchyma: living cells with unevenly thickened corners that give flexibility (1+1).	1+1
27	Sepals, petals, stamens and pistil (2).	2
28	Fraction of radiation reflected (1); snow high, dark forest or ocean low (1).	1+1

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Homogeneous mixtures of metals/metal and non-metal (1); they have properties better than the pure metal (hardness, resistance) (1); brass, bronze (1).	1+1+1
30	U-235: nuclear fuel; Co-60: cancer treatment; I-131: goitre; C-14: dating fossils (any three) (1+1+1).	1+1+1
31	L = √(30² + 40²) = 50 cm (1.5); MA = 50/30 = 1.67 (1.5).	1½+1½
32	ν = 344/3.44 = 100 Hz (1.5); T = 0.01 s (1.5).	1½+1½
33	Single phloem cells were grown in a medium with sugars and hormones (1.5); they dedifferentiated, formed a cell mass and grew into a whole plant, showing totipotency (1.5).	1½+1½

34	Fishes (rohu), amphibians (frog), reptiles (lizard), birds (crow), mammals (human) (3).	3
35	No (1). The gaseous products (CO_2 and water vapour) escape into the air (1), so a closed system would show the same total mass (1).	1+1+1
	OR: Cl (2,8,7) needs one electron; each atom shares one electron, forming one shared pair (1+1); Cl—Cl single bond, each atom now has 8 electrons (1).	1+1+1
36	Radio, microwave, infrared, visible, ultraviolet, X-rays, gamma rays (2); all travel at 3×10^8 m s^{-1} in vacuum (1).	2+1
	OR: Harmful: damages eyes and skin, may cause cancer (1.5); useful: kills germs in water purifiers (1.5).	1½+1½

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) Energy released when nuclei split or combine; used in electricity generation, medicine, research (2). (ii) Raja Ramanna (1). (iii) Heat from nuclear reactions makes steam that drives turbines (2).	2+1+2
	OR: (i) Zinc chloride and hydrogen gas (1). (ii) Hydrogen escaped (1). (iii) Use a closed flask with a balloon or stopper (2). (iv) Equal masses before and after (1).	1+1+2+1
38	(i) Force opposing relative motion between surfaces in contact (1). (ii) Pull a block with a spring balance on surfaces; reading when it just moves is the friction (2). (iii) Foot pushes back, friction pushes forward (2).	1+2+2
	OR: (i) Rate of change of momentum is proportional to net force (1). (ii) Product of mass and velocity (1). (iii) $\Delta p = 10$, $F = 10/0.5 = 20$ N (3).	1+1+3
39	(i) Genetically regulated death of cells (1). (ii) Cancer cells divide without control and spread (2). (iii) To avoid excess cells, tumours and loss of function (2).	1+2+2
	OR: (i) 4 mm ÷ 20 = 0.2 mm = 200 µm (1). (ii) 100X (1). (iii) 200 µm × 100 = 20 000 µm = 20 mm (2). (iv) Resolution of the lenses (1).	1+1+2+1
40	(i) Site, number of eggs, examples (2). (ii) The embryo is protected inside (1). (iii) Fertilisation in a dish, then the embryo placed in the uterus (2).	2+1+2
	OR: (i) Prevents unplanned pregnancy and infections (1). (ii) HIV/AIDS, herpes; use of condoms and avoiding contact with infected persons (2). (iii) Condoms, pills, copper-T with how each works (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER